

PONDICHERRY UNIVERSITY
RAMANUJAN SCHOOL OF MATHEMATICAL SCIENCES
DEPARTMENT OF STATISTICS



M.Sc. QUANTITATIVE FINANCE
MSQF 531: Applied Time Series Analysis and Forecasting
LECTURE NOTES ON

**Applied Time Series Analysis
and Forecasting**

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MSQF 531: APPLIED TIME SERIES ANALYSIS AND FORECASTING

CREDITS: 4

Course Objectives:

- *Providing a clear explanation of the fundamental theory of time series analysis and forecasting.*
- *The course features treatments of forecast improvement with regression and autoregression combination models and model and forecast evaluation, along with a sample size analysis for common time series models to attain adequate statistical power.*

Course Outcomes (COs):

CO1: Understand and interpret the fundamental concepts and components of time series data

CO2: Apply smoothing techniques for real world data using R and Excel

CO3: Distinguish between stationary and non-stationary time series

CO4: Application and modeling time-varying volatility using financial data

CO5: Evaluate and compare various forecasting methods using real time data

Unit I: Introduction to Time Series

Definition and examples of Time Series Models — Graphical Representation of Time Series Data — Approaches of Time Series — Additive and multiplicative approach — Components and various decompositions of Time Series Models — Numerical description of Time Series — Data transformations — Methods of estimation — Trend, Seasonal and exponential.

Unit II: Smoothing Techniques and Univariate Time Series Models

Smoothing Techniques: Moving Averages: Simple, centered, double and weighted moving averages; single and double exponential smoothing — Holt's and Winter's methods — Exponential smoothing techniques for series with trend and seasonality — First and Second order AR and MA Models — ARMA/ARIMA Models — Mixed ARMA/ARIMA models their statistical Properties — Box-Jenkins methodology ACF and PACF functions — Finite order $AR(p)$ and $MA(q)$ models.

Unit III: Multivariate and Multiple Equation Models

Trend stationary — Stationary Time Series Models — General Unit Root Tests: Dickey-Fuller Test, Augmented Dickey-Fuller Test — Johansen Test for cointegration — Granger causality — Error Correction Model (ECM) — Vector Autoregressive (VAR) model — Vector Error Correction Model (VECM), estimation of Lag models.

Unit IV: Modeling Volatility

Impulse response function, variance decomposition — Definition and representation of ARCH and GARCH Models — Their use in financial time series data — Volatility forecasting using GARCH(1, 1) Model — Diagnostic checking of model — Analysis of residuals.

Unit V: Applications and Forecast Evaluation

Introduction to business forecasting — Scope — Types of forecasting — Forecasting cycle — Different forecasting techniques — Exploring data patterns and choosing forecasting technique — Managing forecasting process — Measuring forecasting error — Forecasting error comparison — Forecast accuracy measures (MAE, MSE, RMSE, MAPE) — Model validation and backtesting — Scenario analysis and stress testing in financial forecasting — Case studies: stock prices, interest rates, macroeconomic forecasting — Use of software tools (R, Python) for applied time series modeling.

Books for Reference:

1. Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.).
2. Hamilton, J. D. (2023). *Time Series Analysis* (2nd ed.). Princeton University Press.
3. Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (1994). *Time Series Analysis: Forecasting and Control* (3rd ed.). Prentice Hall.
4. Makridakis, S., & Wheelwright, S. C. (1997). *Forecasting: Methods and Applications* (3rd ed.). John Wiley & Sons.
5. Mills, T. C. (1997). *The Econometric Modelling of Financial Time Series* (2nd ed.). Cambridge University Press.

6. Shumway, R. H., & Stoffer, D. S. (2023). *Time Series Analysis and Its Applications: With R Examples* (4th ed.). Springer.
7. Tsay, R. S. (2024). *Analysis of Financial Time Series* (4th ed.). John Wiley & Sons.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XX	XX	XX	XX
CO2	XXX	XXX	XX	XX	XX
CO3	XXX	XX	XX	XX	XX
CO4	XXX	XXX	XXX	XX	XX
CO5	XXX	XXX	XXX	XX	XXX

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Chapter 1

Introduction to Time Series

1.1 Time Series Definition & Decomposition Models

Definition 1.1: Time Series

A **Time Series** is data which is collected over a period of time and arranged in a chronological order.

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1.1.1 Components of Time Series

A time series Y consists of four primary components:

- $T \rightarrow$ **Trend**: Long-term progression or general direction in the data.
- $S \rightarrow$ **Seasonality**: Short-term periodic fluctuations repeating at regular intervals.
- $C \rightarrow$ **Cyclical**: Wavelike oscillations over longer durations (business/economic cycles).
- $I \rightarrow$ **Irregular Component**: Random, unpredictable variations or noise.

1.1.2 Decomposition Models

The relationship among the four components is represented as:

Decomposition Formulations

1. Additive Model:

$$Y = T + S + C + I \quad (1.1)$$

2. Multiplicative Model:

$$Y = T \times S \times C \times I \quad (1.2)$$

Logarithmic Transformation

Applying the natural logarithm transforms a multiplicative model into an additive form:

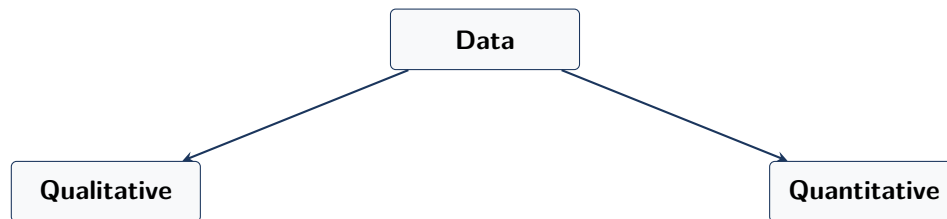
$$Y = S \cdot C \cdot T \cdot I \implies \ln(Y) = \ln(S) + \ln(C) + \ln(T) + \ln(I) \quad (1.3)$$

1.2 Time Series Analysis & Forecasting Fundamentals

1.2.1 Classification of Data

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In time series analysis and forecasting, data is broadly categorized into:



1.2.2 Types of Data

Based on observation structures:

1. Time Series Data
2. Cross-Sectional Data
3. Panel Data

1.2.3 Benefits of Time Series

General benefits include understanding past behaviors, modeling underlying dynamics, forecasting future outcomes, and supporting operational planning.

1.3 Data Notations, Analytical Uses & Forecasting Taxonomy

1.3.1 Mathematical Representation of Data Types

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Depending on the dimensional index of observation:

- **Time Series (TS):** Observations indexed chronologically over time t :

$$\text{TS} = y_t \quad (1.4)$$

- **Cross-Sectional (CS):** Observations across multiple individual entities i at a single point in time:

$$\text{CS} = y_i \quad (1.5)$$

- **Panel Data (P):** Observations across both entities i and time periods t :

$$\text{P} = y_{it} \quad (1.6)$$

Example 1.1: Time Series Sales Realization

An illustration of a univariate time series showing sales (y_t) over time periods $t = 1, 2, \dots, 7$:

Time (t)	Sales (y_t)
1	20
2	16
3	12
4	8
5	4
6	0
7	-4

1.3.2 Analytical Uses & Benefits

- **Descriptive (Time Series):** Describing and tracking temporal movements.
- **Spatial (Cross-Sectional):** Evaluating spatial variations across entities.
- **Explorative:** Analyzing relationships and interactions between two or more variables.

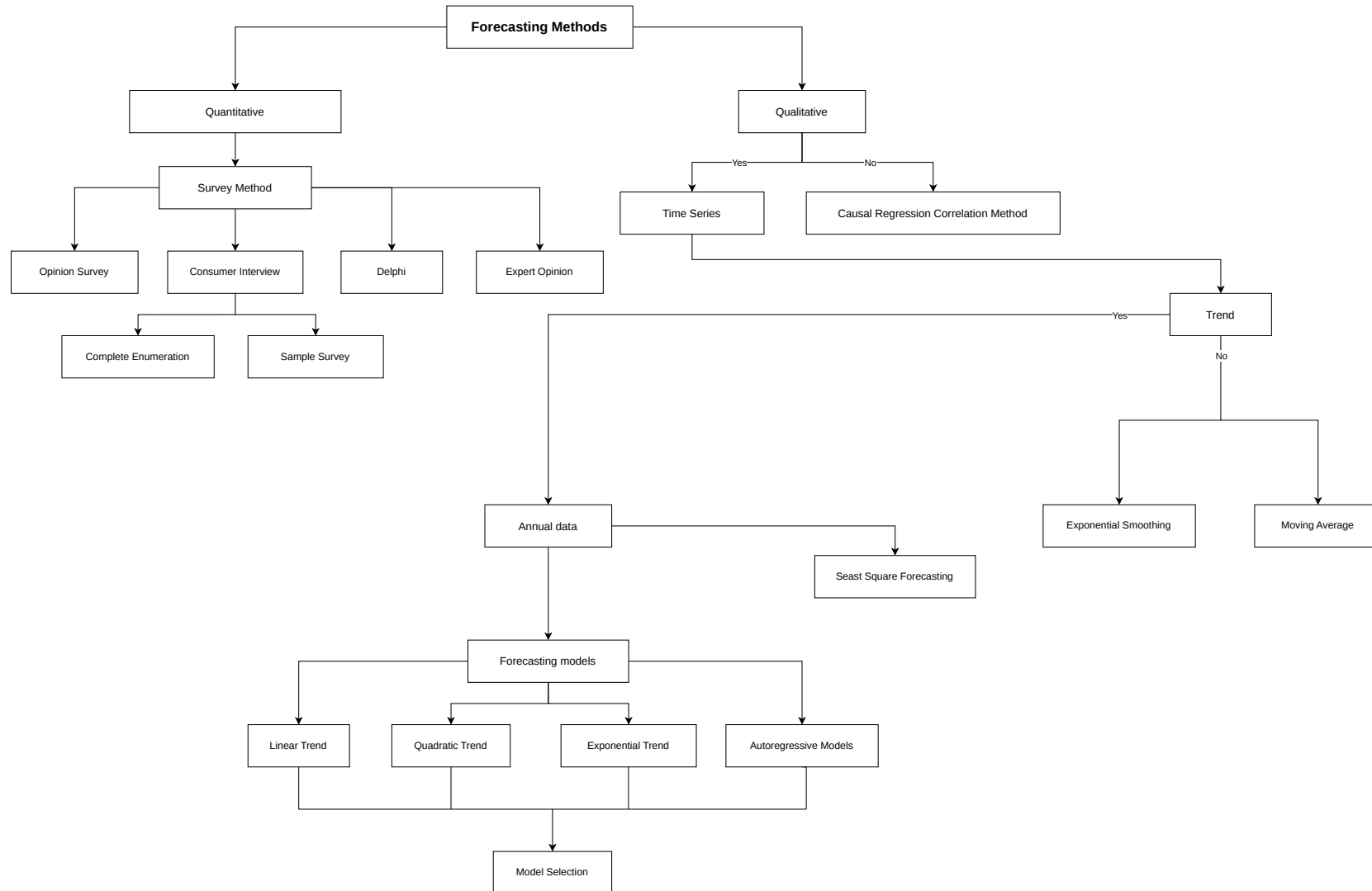
1.3.3 Key Application Areas

1. **Economic:** Macroeconomic trends, inflation, growth indices.
2. **Sales:** Revenue forecasting, sales volume trajectories.
3. **Performance:** Operational efficiency, financial asset returns.
4. **Portfolio:** Asset allocation, risk-return profiling.
5. **Demand:** Consumer demand estimation, inventory planning.
6. **Yield:** Bond yields, interest rates, agricultural outputs.

1.3.4 Forecasting Approaches & Horizons

- **Approaches to Forecasting:**
 - *Quantitative:* Model-based statistical methods on historical numerical series.
 - *Qualitative:* Intuition, expert judgment, and opinion gathering.
 - *Combination (Team / Consensus):* Synthesizing multiple quantitative and qualitative models.
- **Forecasting Horizons (Types):**
 - **Short Term (ST):** Immediate horizon / up to 3 months.
 - **Medium Term (MT):** 3 to 6 months / 1 year.
 - **Long Term (LT):** Extended planning horizons (5+ years).

1.3.5 Comprehensive Taxonomy of Forecasting Methodologies



1.3.5 Trend Models for Annual Data

For annual time series exhibiting long-term movement, trend equations are specified and estimated via least squares or autoregression before model selection:

Summary of Trend Models

1. **Linear Trend:**

$$Y_t = \beta_0 + \beta_1 t + \varepsilon_t \quad (1.7)$$

2. **Quadratic (Curvilinear) Trend:**

$$Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \varepsilon_t \quad (1.8)$$

3. **Exponential Trend:**

$$Y_t = \beta_0 e^{\beta_1 t} \cdot \varepsilon_t \iff \ln(Y_t) = \ln(\beta_0) + \beta_1 t + \ln(\varepsilon_t) \quad (1.9)$$

4. **Autoregressive Trend Model (AR(p)):**

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t \quad (1.10)$$